

ENPH 352

EX1: Skin Depth Prelab

Maxwell's eqns:

$$\begin{aligned} \textcircled{1} \vec{\nabla} \times \vec{H} &= \vec{J} + \epsilon \frac{\partial \vec{E}}{\partial t} \quad \text{where } \vec{J} = \sigma \vec{E} \quad \left. \vphantom{\vec{\nabla} \times \vec{H}} \right\} \vec{H} \text{ field near/in a conductor} \\ \textcircled{2} \vec{\nabla} \times \vec{E} &= -\frac{\partial \vec{B}}{\partial t} = -\mu \frac{\partial \vec{H}}{\partial t} \end{aligned}$$

For an initial field $\vec{H} = H_z \hat{z}$,

→ If nothing varies in time, then:

$$\begin{aligned} \textcircled{1} \vec{\nabla} \times \vec{H} &= \vec{J} \quad \text{where } \vec{\nabla} \times \vec{H} = \left(\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} \right) \hat{x} + \left(\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} \right) \hat{y} + \left(\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \right) \hat{z} \\ &= \frac{\partial H_z}{\partial y} \hat{x} - \frac{\partial H_z}{\partial x} \hat{y} \end{aligned}$$

$$\begin{aligned} \text{so, } \vec{J} = \sigma \vec{E} &= \frac{\partial H_z}{\partial y} \hat{x} - \frac{\partial H_z}{\partial x} \hat{y} \\ \vec{E} &= \frac{1}{\sigma} \left[\frac{\partial H_z}{\partial y} \hat{x} - \frac{\partial H_z}{\partial x} \hat{y} \right] \end{aligned}$$

$$\textcircled{2} \vec{\nabla} \times \vec{E} = 0$$

$$0 = \vec{\nabla} \times \left[\frac{\partial H_z}{\partial y} \hat{x} - \frac{\partial H_z}{\partial x} \hat{y} \right] \neq$$

$$0 = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial H_z}{\partial y} & -\frac{\partial H_z}{\partial x} & 0 \end{vmatrix}$$

$$0 = \frac{\partial}{\partial z} \left(\frac{\partial H_z}{\partial x} \right) \hat{x} - \frac{\partial}{\partial z} \left(\frac{\partial H_z}{\partial y} \right) \hat{y} + \left(-\frac{\partial^2 H_z}{\partial x^2} - \frac{\partial^2 H_z}{\partial y^2} \right) \hat{z}$$

$$\boxed{\vec{\nabla} \times \vec{E} = 0 \text{ only if } \frac{\partial H_z}{\partial x} = 0 \checkmark}$$

→ If $\sigma = 0$ and no $\vec{J}$, $\vec{E} \perp \vec{H}$ and travel as a wave:

$$\textcircled{1} \vec{\nabla} \times \vec{H} = \epsilon \frac{\partial \vec{E}}{\partial t}$$

Prove that it can be expressed as wave eqn: $\nabla^2 \vec{H} = \mu \epsilon \frac{\partial^2 \vec{H}}{\partial t^2}$

Use vector identities: $0 \rightarrow$ no magnetic monopoles

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{H}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{H}) - (\vec{\nabla} \cdot \vec{\nabla}) \vec{H}$$

$$\vec{\nabla} \times \left(\epsilon \frac{\partial \vec{E}}{\partial t} \right) = -\nabla^2 \vec{H}$$

$$\epsilon \frac{\partial}{\partial t} (\vec{\nabla} \times \vec{E}) = -\nabla^2 \vec{H}$$

$$\textcircled{2} \epsilon \frac{\partial}{\partial t} \left(-\mu \frac{\partial \vec{H}}{\partial t} \right) = -\nabla^2 \vec{H}$$

$$\boxed{\nabla^2 \vec{H} = \mu \epsilon \frac{\partial^2 \vec{H}}{\partial t^2} \checkmark \text{ wave eqn}}$$

$$\textcircled{2} \vec{\nabla} \times \vec{E} = -\mu \frac{\partial \vec{H}}{\partial t}$$

Prove $\vec{E}$ can also be expressed as wave eqn: $\nabla^2 \vec{E} = \mu\epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$

Use vector identities: $0 \rightarrow$ in free space

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = \nabla(\vec{\nabla} \cdot \vec{E}) - \nabla^2 \vec{E}$$

$$\vec{\nabla} \times (-\mu \frac{\partial \vec{H}}{\partial t}) = -\nabla^2 \vec{E}$$

$$-\mu \frac{\partial}{\partial t} (\underbrace{\vec{\nabla} \times \vec{H}}_0) = -\nabla^2 \vec{E}$$

$$\mu\epsilon \frac{\partial}{\partial t} (\frac{\partial \vec{E}}{\partial t}) = \nabla^2 \vec{E}$$

$$\boxed{\nabla^2 \vec{E} = \mu\epsilon \frac{\partial^2 \vec{E}}{\partial t^2} \checkmark \text{ wave eqn}}$$

Because both $\vec{E}$ and $\vec{H}$ satisfy the wave eqn in free-space, we know they travel as a single electromagnetic wave where $\vec{E} \perp \vec{H} \perp \hat{k}$ (wave vector).
Can also verify by checking if $\vec{E} \cdot \vec{H} = 0$ for a simple plane wave.

Finding dispersion relation:

Assume $\vec{E}$ and $\vec{H}$ are harmonic where $\vec{H} = H_z \exp[i(k_x x - \omega t)] \hat{z}$
which means $\vec{E} = E_y \exp[i(k_x x - \omega t)] \hat{y}$ also propagates along $\hat{x}$ but oscillates in $\hat{y}$.

$\rightarrow$ Taking cross-products of $\vec{E}$ and $\vec{H}$:

$$\begin{aligned} \textcircled{1} \vec{\nabla} \times \vec{H} &= \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 0 & H_z \exp \dots \end{vmatrix} \\ &= \frac{\partial}{\partial y} (H_z \exp \dots) \hat{x} + \frac{\partial}{\partial x} (H_z \exp \dots) \hat{y} \\ &= \frac{\partial}{\partial x} [H_z \exp(i(k_x x - \omega t))] \hat{y} \\ &= i k_x H_z \exp(i(k_x x - \omega t)) \hat{y} \end{aligned}$$

$$\begin{aligned} \textcircled{2} \vec{\nabla} \times \vec{E} &= \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & E_y \exp \dots & 0 \end{vmatrix} \\ &= \frac{\partial}{\partial z} (-E_y \exp \dots) \hat{x} + \frac{\partial}{\partial x} (E_y \exp \dots) \hat{z} \\ &= \frac{\partial}{\partial x} [E_y \exp(i(k_x x - \omega t))] \hat{z} \\ &= i k_x E_y \exp(i(k_x x - \omega t)) \hat{z} \end{aligned}$$

→ Now, expanding $\vec{E}$ and $\vec{H}$ in Maxwell eqns ① and ②

$$\begin{aligned} \textcircled{1} \quad \vec{\nabla} \times \vec{H} &= \vec{J} + \epsilon \frac{\partial \vec{E}}{\partial t} \\ &= \sigma \vec{E} + \epsilon \frac{\partial}{\partial t} [E_y \exp(i(k_x x - \omega t)) \hat{y}] \\ &= \sigma E_y \exp(i(k_x x - \omega t)) \hat{y} - i\omega \epsilon E_y \exp(i(k_x x - \omega t)) \hat{y} \\ &= E_y \exp(i(k_x x - \omega t)) (\sigma - i\omega \epsilon) \hat{y} \end{aligned}$$

$$\begin{aligned} \textcircled{2} \quad \vec{\nabla} \times \vec{E} &= -\mu \frac{\partial \vec{H}}{\partial t} \\ &= -\mu \frac{\partial}{\partial t} [H_z \exp(i(k_x x - \omega t)) \hat{z}] \\ &= -i\omega \mu H_z \exp(i(k_x x - \omega t)) \hat{z} \end{aligned}$$

→ Equating both eqns for ① and ②:

$$\begin{aligned} \textcircled{1} \quad \vec{\nabla} \times \vec{H} &= i k_x H_z \exp(i(k_x x - \omega t)) \hat{y} = E_y \exp(i(k_x x - \omega t)) (\sigma - i\omega \epsilon) \hat{y} \\ i k_x H_z &= E_y (\sigma - i\omega \epsilon) \quad \textcircled{3} \end{aligned}$$

$$\begin{aligned} \textcircled{2} \quad \vec{\nabla} \times \vec{E} &= i k_x E_y \exp(i(k_x x - \omega t)) \hat{z} = -i\omega \mu H_z \exp(i(k_x x - \omega t)) \hat{z} \\ k_x E_y &= -\omega \mu H_z \quad \textcircled{4} \end{aligned}$$

→ Find relationship between k^2 and ω^2 :

Multiply ③ and ④:

$$\begin{aligned} i k_x^2 H_z E_y &= -E_y H_z \omega \mu (\sigma - i\omega \epsilon) \\ i k_x^2 &= -\omega \mu \sigma + i\omega^2 \mu \epsilon \end{aligned}$$

$$\boxed{k_x^2 = -\omega^2 \mu \epsilon + i\omega \mu \sigma \text{ complex only if } \sigma \neq 0} \quad \checkmark$$

→ Solve for k_x and penetration depth:

Got some help from TA but seems like we can assume $\sigma \gg \omega \epsilon$ for a good conductor... which means

$$\begin{aligned} k_x^2 &\approx i\omega \mu \sigma \\ &= e^{i\frac{\pi}{2}} \sqrt{\omega \mu \sigma} \\ k_x &= e^{i\frac{\pi}{4}} \sqrt{\omega \mu \sigma} \\ &= \sqrt{\omega \mu \sigma} \left(\sin \frac{\pi}{4} + i \cos \frac{\pi}{4} \right) \end{aligned}$$

$$\sin \frac{\pi}{4} = \cos \frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

$$\boxed{k_x = \underbrace{\sqrt{\frac{\omega \mu \sigma}{2}}}_{Re} + i \underbrace{\sqrt{\frac{\omega \mu \sigma}{2}}}_{Im}}$$

Skin depth is defined as when $\vec{H}$ drops to $\sim 37\% = e$ of its surface value.

For $\vec{H} = H_z \exp \left[i x \left(\sqrt{\frac{\omega \mu_0}{2}} + i \sqrt{\frac{\omega \mu_0}{2}} \right) - i \omega t \right] \hat{z}$

$$= H_z \exp \left[\left(i \sqrt{\frac{\omega \mu_0}{2}} - \sqrt{\frac{\omega \mu_0}{2}} \right) x - i \omega t \right] \hat{z}$$

$$= H_z \exp \left(i \sqrt{\frac{\omega \mu_0}{2}} x \right) \underbrace{\exp \left(-\sqrt{\frac{\omega \mu_0}{2}} x \right)}_{\substack{\text{this is not} \\ \text{sinusoidal,} \\ \text{it's an exponential} \\ \text{decay}}} \exp(-i \omega t) \hat{z}$$

Imaginary part of k_x dictates the penetration/skin depth which happens when $x = \sqrt{\frac{2}{\omega \mu_0}}$.

skin depth: $\delta = \frac{1}{\text{Im}(k_x)} = \sqrt{\frac{2}{\omega \mu_0}} = \sqrt{\frac{1}{\pi f \mu_0}}$

$\swarrow$ freq $\swarrow$ permeability $\swarrow$ conductivity

